PEARL AYEM

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KEY ACHIEVEMENTS

- **Climate data engineering at scale:** Re-designed and built a 680 TB planetary climate pipeline, cutting cloud storage costs by 94% and runs from days to ∼14 hours.
- **Commercial growth:** Personally closed climate risk contracts worth up to £820K. Advised C-suites and boards of companies with over $200 billion in combined revenue.
- **Policy at the highest level:** Designed the data model for the Net-Zero Data Public Utility, now hosted by the UNFCCC with data on 10,000+ companies. Wrote speech content delivered by Mark Carney and Michael Bloomberg, and worked alongside President Macron's office.
- **Cross-functional leadership:** Lead teams of economists, engineers, data scientists and finance specialists. Led 40 analysts for GFANZ's COP27 data launch.

EXPERIENCE

Lead Environmental Risk Modeller —— Risilience *Mar 2025 to Present*

- Lead a team of modellers, economists, data scientists and engineers, quantifying the financial risk of drought, freeze, floods, agricultural yield loss and nature depletion across clients' global supply chains.
- Build geospatial digital twins of the assets and facilities of some of the world's biggest food, fashion and entertainment brands including Nestlé, Mondelez, Barry Callebaut, Burberry, Primark, Inditex, H&M and Lululemon..
- Build parametric and stochastic models climate catastrophe, yield and geopolitical risk across portfolios for asset managers, private equity firms, insurers and underwriters.
- Design models based on first principles atmospheric physics, with bias-corrected CMIP6 projections, and multivariate inputs to calculate future climate risk (case study).
- Turn climate science into financial decisions: projected yield losses on raw materials feed clients' sourcing, supply-chain and financial planning with multi-scenario models upto the year 2100.
- Architected model to product cloud pipeline on BigQuery, processing over 570 billion rows of climate data.
- Rewrote daily simulation as JavaScript UDFs in BigQuery's V8 engine. Flat, streamed arrays cut memory per worker from over 512 MB to under 2 MB, so millions of time series run in parallel without crashes.
- Eliminated a 62 TB shuffle spill with hierarchical geohash batching and re-clustered tables, so spatial queries skip over 95% of storage. Reading weather once for all crops cut data scanned by 89%.
- Cut storage from 680 TB to 46 TB, costs by 94%, and full runs from 4 days to ∼14 hours.
- Lead the firm's AI work on raw material risk, built agentic workflows (LangGraph, LangSmith) to add new crops to existing models, and halved model crop release cycles, from 8 weeks to 4.
- Close around two contracts a year (£150K to £200K) and support six more RFPs. Shape the product roadmap with the Head of Product and lead model releases with clients.

Senior Climate Risk Modeller —— Sillion *Dec 2023 to Feb 2025*

- Strategic and technical lead in a boutique transition risk consultancy, choosing which sectors and products to pursue.
- Identified climate risk to data centres and electrification supply chains ahead of the AI investment boom, through market research and climate trend analysis.
- Research led to closing contracts for climate risk to data centre operations with Schneider Electric, and copper component pricing and supply for RS Components.
- Sold to asset managers, banks, insurers and construction firms, closing contracts worth up to £820K.
- Built transition plans, SBTi strategy, and risk KPIs for boards of brands like Reckitt, Aston Martin and Keller construction.
- Turned risk models into board decisions: bespoke dashboards and earnings-at-risk models that fed clients' TCFD/ISSB, CSRD and TPT disclosures and annual reports.
- Built the firm's first transition-planning dashboard and led product for an ESG data tool tracking 600+ companies.
- Mentored analysts and consultants on building transition models, market research and data aggregation for financial risk.
- Keynote panellist at Chartis Research's ClimateRisk50 2024 launch, alongside HSBC.
- Featured in the British Science Association's Smashing Stereotypes (2025) and Cambridge's Net Zero by 2040 series.

Net-Zero Data Lead —— Bloomberg L.P. (secondment to GFANZ) *Jun 2022 to Nov 2023*

- Designed the data model for the Net-Zero Data Public Utility, launched at COP28.
- Ran vendor selection with the CTO and helped write the RFP. Oversaw Google's engineers through to launch.
- Built ETL pipelines and APIs ingesting data from hundreds of providers.
- Helped secure the CDP partnership, bringing 10,000+ companies onto the platform.

- Led 40 analysts for GFANZ's COP27 data launch.
- Wrote speech content delivered by Mark Carney and Michael Bloomberg.
- Worked with President Macron's office, the governments of Switzerland and Singapore, CDP and the UNFCCC.
- Sold climate data to buy-side clients, including Vanguard and Citadel Commodities.
- Co-authored two reports with partners from the ISSB and IEA.

Climate Science Researcher —— University of Cambridge & British Antarctic Survey *Oct 2021 to Jun 2022*

- Fully funded research on the atmospheric drivers of Antarctic surface melt.
- Applied a CNN and EOF analysis to 1.7 billion geospatial data points. The work is now MPhil teaching material.

Early career *2016 to 2021*

- **Geospatial Engineer, UAViation Aerial Solutions:** led LiDAR drone mapping across BC, Alberta and Yukon. Delivered 18 projects in a year and led the software team. The company was later acquired by Volatus Aerospace.
- **Research Assistant, UBC Atmospheric Science:** installed weather stations to WMO standards and took atmospheric measurements during wildfires. Built satellite data pipelines on AWS.
- **Lead Teaching Assistant, UBC Computer Science:** managed 18 TAs and taught Python and C++ to 1,000 students. Best TA three years running.
- **Sustainability Brand Manager, P.G.A. Hospitality:** raised CAD 1.5M in investment and grew revenue by 233%.

EDUCATION

University of Cambridge, MPhil in Polar Studies (Atmospheric Physics) *2021 to 2022*

- Scott Polar Scholarship: full scholarship awarded to the top applicant.

University of British Columbia, BSc in Atmospheric Science *2015 to 2020*

- Graduated top of class. Dean's Honour Roll, 2017 to 2020.

PUBLICATIONS

- *Recommendations and Guidance on Financial Institution Net-Zero Transition Plans* (co-author). GFANZ, November 2022.
- *Development of the Net-Zero Data Public Utility* (co-author). NZDPU and Bloomberg, November 2022.
- *Request for Proposal: Net-Zero Data Public Utility* (contributor). NZDPU and Bloomberg, November 2022.

BOARD ROLES

- **Board Member,** Compassionate Eye Foundation (2024 to 2026). Led revenue strategy for funding projects.
- **Co-Organiser,** Polar Impact (2021 to 2023). Won £50,000 in grants and ran COP26 climate justice panels.

SELECTED CERTIFICATIONS

Data Engineer in Python (DataCamp, 2024) ▪ Bloomberg Market Concepts (2023) ▪ Corporate Finance Foundations (LinkedIn, 2022)

SKILLS

Leadership:	Team leadership and mentoring; strategy and product roadmaps; RFPs and sales; board and C-suite advisory; translating between scientists, engineers and executives
Languages:	Python, SQL, JavaScript, C++
Cloud:	Google Cloud (BigQuery, Vertex AI, Cloud Build); AWS (S3, EC2); Docker
Data Engineering:	PySpark, Kafka; pandas, Pandera, pydantic; Parquet, Zarr; ETL/ELT, orchestration, data quality; BigQuery optimisation (clustering, UDFs, cost)
CI/CD & Delivery:	Git, GitHub Actions, GitLab CI, Cloud Build; Figma for collaborating with product and front-end teams
Geospatial & Climate:	xarray, geopandas, rasterio, Shapely, GDAL, netCDF/GRIB; CMIP6, bias correction, downscaling;
ML & AI:	CNNs, PCA, EOFs; LangGraph, LangSmith, Vertex AI
Risk & Standards:	Physical, transition and supply-chain risk; earnings-at-risk; ISSB, TCFD, CSRD, TPT, TNFD, CDP; OSFI B-15, CSDS 1 and 2